Title of Final Project: Are You Actually Replaceable? What the Data Says About AI and Your Career

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Goal (or Thesis)

This project tests whether occupations already flagged as highly "exposed" to AI using the AIOE (AI Occupational Exposure) score from Felten, Raj, and Seamans actually show the wage and employment shifts that AI-displacement narratives predict. We will link each occupation's AIOE score to its BLS OEWS employment/wage figures by SOC code, then run a linear regression (with a skill/education interaction term) to test whether AIOE predicts real outcomes, and whether that relationship differs for high- versus low-skill occupations. The interaction term matters because the AI-displacement narrative is not just "high exposure means job loss", it is often specifically that high-skill, high-exposure roles behave differently from low-skill, high-exposure ones, so a model without that interaction would blur two potentially opposite effects into one misleading average. Before any of that modeling can happen, this phase of the project is about getting both source datasets fully understood and cleaned, since a regression built on unexamined or badly formatted data would not be trustworthy evidence either way.

Description of the Datasets

Size of the Datasets

- Dataset 1 — BLS OEWS (national_M2025_dl.xlsx): 1,401 rows × 32 columns raw; 28 columns after cleaning. Every row has AREA_TITLE = "U.S." and NAICS_TITLE =

"Cross-industry," confirming this is the national, all-industries cut of OEWS which is exactly the scope the research question needs, since the plan is to join on SOC occupation code rather than on geography or industry.

- Dataset 2 — AIOE Appendix A (AIOE_DataAppendix.xlsx): 774 rows × 3 columns, unchanged after cleaning (none was needed). One AIOE score per SOC occupation. The AIOE workbook ships five appendices total, but only Appendix A is used here, since the other four break exposure down by industry or geography, which this project does not need.

Together, the two files are already the right shape for the eventual merge since both are keyed on SOC-style occupation codes rather than on geography, industry, or time , but Dataset 1 is roughly 1.8× larger than Dataset 2, which foreshadows the join coverage question addressed later in Preliminary Statistical Analysis.

Column Names

Dataset 1 (28 cleaned columns): AREA, AREA_TITLE, AREA_TYPE, PRIM_STATE, NAICS, NAICS_TITLE, I_GROUP, OWN_CODE, OCC_CODE, OCC_TITLE, O_GROUP, TOT_EMP, EMP_PRSE, MEAN_PRSE, ANNUAL, HOURLY, H_MEAN, H_PCT10, H_PCT25, H_MEDIAN, H_PCT75, H_PCT90, A_MEAN, A_PCT10, A_PCT25, A_MEDIAN, A_PCT75, A_PCT90. (JOBS_1000, LOC_QUOTIENT, PCT_TOTAL, and PCT_RPT were part of the raw 32-column file but dropped during cleaning ; see Data Wrangling.)

Dataset 2 (3 columns): SOC Code, Occupation Title, AIOE.

Classification of Each Column

Column(s)	Type	Notes
AREA_TITLE, NAICS_TITLE, I_GROUP, OWN_CODE, etc.	Categorical (nominal)	Constant — national/cross-industry cut
OCC_CODE, OCC_TITLE / SOC Code, Occ. Title	Categorical (nominal)	6-digit SOC ID + name, ~unique per row

Column(s)	Type	Notes
O_GROUP	Categorical (ordinal)	total → major → minor → broad → detailed
ANNUAL, HOURLY	Categorical (binary flag)	TRUE / blank for annual- or hourly-only pay
TOT_EMP	Quantitative (discrete)	Headcount, rounded to nearest 10
EMP_PRSE, MEAN_PRSE	Quantitative (continuous)	Percent relative std. error
H_MEAN–H_PCT90	Quantitative (continuous)	Hourly wage, USD/hr
A_MEAN–A_PCT90	Quantitative (continuous)	Annual wage, USD/yr
AIOE	Quantitative (continuous)	Standardized (z-score) exposure index

The constant nominal columns (AREA_TITLE, NAICS_TITLE, etc.) are still worth classifying rather than dropping outright: keeping them confirms, rather than assumes, that the file is the single national/cross-industry cut it claims to be, which is the sanity check the join in a later phase will depend on. O_GROUP is the one ordinal column in either dataset, its five levels form a real hierarchy (total contains major, which contains minor, and so on down to detailed), which matters for the merge: only detailed-level rows should be joined to AIOE's occupation-level scores, since joining at a coarser level would double-count the more granular AIOE occupations underneath it.

List of Discrete Values for Categorical Variables

- O_GROUP: total, major, minor, broad, detailed (the 5 SOC hierarchy levels).
- ANNUAL, HOURLY: True / False.

- All other categorical columns are either constant across the file or effectively unique identifiers (OCC_CODE/SOC Code, occupation titles), so no meaningful, small category list applies to them.

Distribution of Discrete Variables

O_GROUP's mode is detailed, with 830 of 1,401 rows (about 59%); counts fall off sharply through broad, minor, major, total. This is expected, not a data-quality concern: BLS reports one total row and a handful of major/minor groups, then progressively more granular occupations as the hierarchy gets finer, so "detailed" being both the mode and the majority of rows simply reflects that detailed occupations are the base unit of the classification. It also means that once the two datasets are merged, the working sample for the regression will be built almost entirely from these ~830 detailed rows, not the higher-level aggregates. SOC Code and Occupation Title in Dataset 2 are unique per row (774 of 774), so no categorical mode applies there , the only thing to confirm is uniqueness, which is held. ANNUAL is False for 1,320 of 1,401 rows and HOURLY is False for 1,394 of 1,401, so both flags are dominated by "does not apply," consistent with most occupations being reported on both an hourly and an annual basis rather than just one.

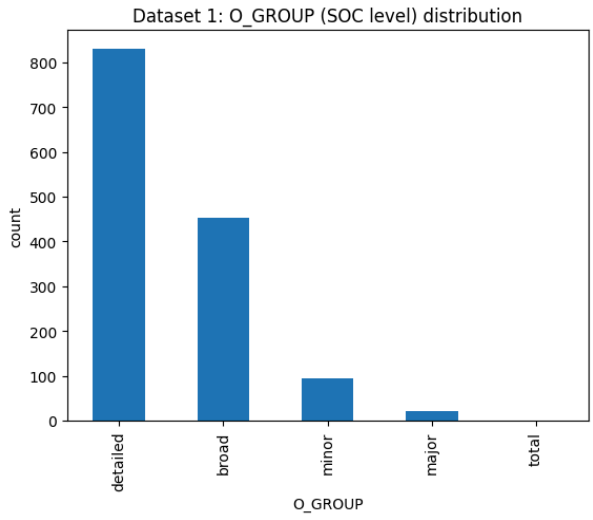


Figure 1. Dataset 1 — rows by O_GROUP. "Detailed" is the mode.

Distribution of Quantitative Variables

Column	Min	Median	Mean	Std. Dev.	Max
TOT_EMP	220	69,990	554,924.93	4,342,791.20	155,495,730
A_MEAN ($)	31,980	65,450	78,378.27	46,831.66	502,050
A_MEDIAN ($)	30,890	60,160	72,615.27	46,327.29	559,030
H_MEAN ($)	15.37	30.59	37.15	22.60	241.37
AIOE (z-score)	-2.6698	-0.0512	≈0	1.0000	1.5277

The wage and employment columns all show mean noticeably above median, for A_MEDIAN, a mean of $72,615 against a median of $60,160, which shows a noticeable right skew: a small number of very high-paying, or in TOT_EMP's case very high-headcount, occupations pull the average up well past where most occupations actually sit. TOT_EMP's skew is extreme (std. dev. of 4.3M against a max of 155.5M, almost entirely driven by the "All Occupations" grand-total row), which is a practical warning for later phases: that single row is a total, not a peer occupation, and needs to be excluded before any regression or correlation analysis, or it will dominate the fit. Wage columns are far better behaved but still right-skewed in the ordinary sense that a few executive/specialist occupations pay much more than the bulk of the workforce, which is why median (not mean) is the more representative "typical wage" figure to lead with when summarizing this dataset. AIOE, by contrast, is standardized to mean ≈ 0 and std ≈ 1 as designed, but Figure 3 shows it is not symmetric, it is left-skewed with a second cluster of occupations bunched up near its top end, meaning there is a distinct group of occupations the AIOE authors rated as highly exposed to AI, rather than exposure trailing off smoothly. That cluster is the group the regression ultimately cares about most, since it is where the AI-displacement hypothesis makes its strongest prediction.

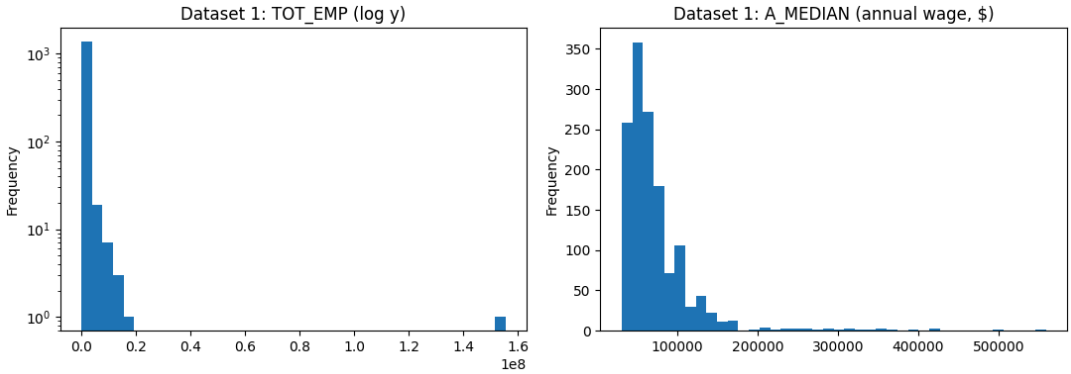


Figure 2. TOT_EMP (log y) and A_MEDIAN distributions.

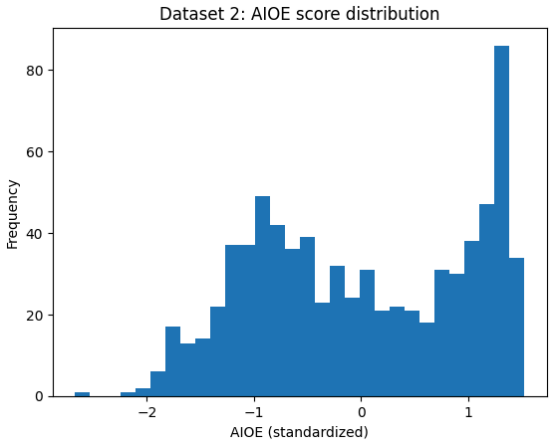


Figure 3. AIOE score distribution.

Data Wrangling

List of Missing Data

Column(s)	Missing	Reason
JOBS_1000, LOC_QUOTIENT, PCT_TOTAL, PCT_RPT	100%	Only apply to state/MSA/industry files, not this national cut
ANNUAL, HOURLY	1,320 / 1,394	Blank = flag doesn't apply, not unknown

Column(s)	Missing	Reason
Wage columns (H_*, A_*)	81 / 7 (via '*' code)	Occupation not reported on that pay basis

Dataset 2 had no missing values in any column, which is a useful cross-check on Dataset 1: it confirms the missingness above is a feature of how OEWS reports national data, not a general data-quality problem with occupational datasets. Every missing pattern in Dataset 1 lines up with a documented BLS convention rather than an unexplained gap and the four fully-empty columns are simply inapplicable at national scope, the ANNUAL/HOURLY blanks encode a real "does not apply" fact, and the wage suppressions mark a pay basis an occupation genuinely is not reported on. That distinction is the basis for the wrangling decisions below: none of this is missing data in the sense of "we don't know the value," so none of it should be treated the way unknown data would be.

Replacement of Missing Data

No missing values were replaced with the mode or the mean anywhere in either dataset. Since each missing value above reflects a documented structural fact rather than an unmeasured gap, imputing a mode or mean would manufacture a number for something that was never meant to exist, for example, giving a suppressed hourly wage a fabricated "average" value would misrepresent an occupation that BLS deliberately did not survey on that basis. Instead, two targeted recodes were applied: ANNUAL/HOURLY blanks were recoded to False rather than left blank, since blank genuinely means "flag does not apply" and treating it as a proper boolean makes the column usable in later filtering or modeling; and the wage columns' '*' suppression codes were converted to NaN and left missing (not imputed), preserving their "not measured on this basis" meaning. Dataset 2 required no replacement, since it had no missing data to begin with.

Deletion of Rows and Reasons

No rows were deleted from either dataset. The four structurally empty columns (JOBS_1000, LOC_QUOTIENT, PCT_TOTAL, PCT_RPT) were dropped from Dataset 1 as columns, not rows, removing rows instead would have thrown away real occupation records purely because of columns that were never going to be populated at this national scope, which would have

needlessly shrunk the sample going into the regression. Seven duplicate OCC_CODE pairs (one broad-level and one detailed-level row for the same occupation) were likewise kept rather than removed, since each pair is two distinct, legitimate BLS estimates at different levels of aggregation, not a data-entry duplicate; dropping either row would have silently discarded a real estimate, while keeping both preserves the ability to choose the right granularity (detailed) when the merge happens.

Check That the Data Is in the Correct Range

Check	Result
TOT_EMP negative values	0
Wage columns negative values	0
EMP_PRSE, MEAN_PRSE outside [0, 100]	0
AIOE values with $\|z\| > 4$ (implausible)	0

Every range check passed with zero violations: no negative employment or wages in Dataset 1, and no implausible relative-standard-error or standardized-AIOE values in either dataset. No rows had to be dropped or corrected on range grounds, so the sample size going into later phases is not reduced by this step.

Check That the Data Is in the Correct Format

Check	Result
OCC_CODE / SOC Code not matching SOC ‘DD-DDDD’ format	0
TOT_EMP not rounded to the nearest 10	0

Every OCC_CODE and SOC Code matches the required SOC 2-digit–hyphen–4-digit format, and every TOT_EMP value is rounded to the nearest 10 exactly as BLS documents — nothing needed a fix in either dataset.

Check for Duplicates and Removal

Check	Result
Dataset 1: fully duplicated rows	0
Dataset 1: OCC_CODE values appearing more than once	7 (kept — see below)
Dataset 2: fully duplicated rows	0
Dataset 2: duplicated SOC Code values	0

Neither dataset contained a true duplicate row. The only repeated identifiers were the 7 OCC_CODE values in Dataset 1 discussed above (one broad-level and one detailed-level row per occupation), which were intentionally kept rather than removed since both rows are legitimate, distinct BLS estimates. A completely clean result on checks this basic — no negative wages, no out-of-range percentages, correctly formatted codes, no true duplicates — is itself a form of evidence that these are well-maintained, professionally produced government and academic datasets rather than scraped or self-reported data, which supports treating their values as reliable inputs to a regression rather than needing further validation.

Preliminary Statistical Analysis

Before computing correlations, we restricted the BLS data to detailed occupations, which removes the "All Occupations" total row and the major, minor, and broad group rows, and also resolves the 7 duplicated occupation codes, since only one row in each pair is detailed-level. We then merged with the AIOE dataset on SOC code. Of the 830 detailed occupations, 673 (about 81%) matched an AIOE score; 157 BLS occupations and 101 AIOE occupations went unmatched, most likely because the AIOE data uses the older 2010 SOC codes while the 2025 BLS data uses the 2018 codes. Missing wage values were left blank rather than imputed, consistent with our wrangling decisions, and were excluded pairwise when computing correlations.

We computed Pearson correlation coefficients among AIOE, total employment, and annual wage measures (median, mean, 10th and 90th percentile). AIOE showed a moderate positive

correlation with annual median wage (r = 0.45), with similar values for mean wage (r = 0.47) and the 10th and 90th percentile wages (r = 0.45 and 0.44), indicating that occupations more exposed to AI tend to be higher-paying. In contrast, AIOE was essentially uncorrelated with total employment (r = 0.02). The wage measures were strongly correlated with one another (r = 0.73–0.97), which is expected since they describe the same underlying pay distribution.

For AIOE and annual median wage, we compared linear, quadratic, and cubic fits. The linear model had an adjusted R^2 of 0.200, the quadratic 0.200, and the cubic 0.199. Because the higher-degree models offer no improvement, the linear fit is the best choice. Results for mean wage were similar (adjusted R^2 of 0.223 linear vs. 0.226 for both higher-degree models). For AIOE and employment, all fits had adjusted R^2 near zero, confirming no meaningful relationship.

These results suggest that AI exposure is concentrated in higher-wage, likely higher-skill occupations, which supports examining skill level as an interaction term in our model. Because this analysis uses a single year (2025), it measures levels rather than changes; testing actual displacement will require an earlier year of OEWS data.

Machine Learning Models

To test the core hypothesis that AI exposure (AIOE) predicts real wage and employment shifts, and whether this effect differs across skill levels, we need models that explicitly handle continuous target outcomes such as wage and employment, while accommodating interaction terms and non-linearities. Some of the machine learning models that can be used to test our hypothesis are listed below

- Ordinary Least Squares Linear Regression with Interaction Terms

  Ordinary Least Squares linear regression is the foundational hypothesis testing model for this project. Because the thesis directly challenges the idea that AI simply replaces workers, OLS helps us test whether AI's impact depends on a worker's skill level. In this framework, the dependent variable is the longitudinal percentage change in median annual wage or total employment between OEWS

survey years, and is modeled using a standardized AIOE exposure z-score which is an indicator for occupation skill level, and an explicit multiplicative interaction term (AIOE×Skill). The main advantage of OLS is its clear interpretability. It provides individual coefficients alongside standard errors, t-statistics, and p-values. Specifically, the interaction coefficient directly shows whether high and low skill occupations experience statistically different wage or employment shifts when exposed to AI. OLS also allows us to check key statistical assumptions, like whether our errors are normally distributed. This creates a clear baseline before we try more complex machine learning models.

- Random Forest Regressor

  The Random Forest regressor builds on the OLS baseline by dropping linear limits and adding non parametric flexibility. As an ensemble model that builds hundreds of decision trees on bootstrapped samples of our occupational data, it captures complex, nonlinear patterns without needing manual feature engineering or pre-set equations. In job displacement, AI exposure rarely follows a straight line because impacts might only trigger after passing a critical threshold, or they may level off past a certain wage ceiling. Random Forest handles these step function patterns naturally through recursive hierarchical splitting. By averaging predictions across many trees, it cuts single tree variance and avoids overfitting all our matched detailed occupations. Finally, feature importance metrics such as mean decrease in impurity and permutation importance let us check whether an occupation's AIOE score truly predicts job market trends better than baseline economic variables like initial median pay or total headcount.

- Gradient Boosting Regressor

  Extreme Gradient Boosting (XGBoost) paired with SHapley Additive explanations (SHAP) delivers the pipeline's highest predictive accuracy and visual clarity. Unlike Random Forest, which builds trees side by side in parallel, XGBoost builds decision trees one after another, continually training each new tree to correct the errors of the previous ones. This

framework consistently yields top performance on structured tabular data like our BLS and AIOE crosswalk. Because complex boosted models can act like black boxes, pairing XGBoost with SHAP values provides complete interpretability, both globally across the dataset and locally for individual occupations. By applying game theory, SHAP calculates the exact contribution of each variable and its interactions to an occupation's predicted wage or employment change. Using SHAP dependence and interaction plots, we can visually map how AI exposure interacts with skill levels across the whole workforce, providing clear, factual evidence of where AI exposure leads to job growth, or stagnation.

Narration

- “Are you actually replaceable?” Every week there’s a new headline saying AI is coming for our jobs.
- We’re three Stats and Data Science students about to enter that job market, so we wanted to know: what does the data actually say?
- Most predictions are about what AI could do, not about what the numbers show
- The usually assumption is that AI will hit routine, lower-paid work first, the jobs that seem easiest to automate
- We decided to test that assumption against real data
- The first source is the Bureau of Labor Statistics' 2025 wage and employment survey, which covers the whole U.S. workforce by occupation.
- The second is the AI Occupational Exposure score from Felten, Raj, and Seamans, a peer-reviewed measure of how much each job's abilities overlap with what AI can do.
- We matched them by occupation code and ended up with 673 occupations, each with an exposure score, a wage, and a headcount.
- Our first question was does exposure predict how many people do a job? We expected to see something here, and we didn't. The line is flat. The correlation is 0.02, which is essentially zero.
- Heavily exposed jobs and barely exposed jobs employ similar numbers of people. Exposure alone doesn't tell you whether a job is big or small.

- The surprise here though, is who is actually exposed? When we plotted exposure against pay, a clear pattern appeared. The line actually goes up. The line goes *up*. The more exposed an occupation is to AI, the more it tends to pay, with a correlation of 0.45.
- Exposure isn't spread evenly. There's a distinct cluster of highly exposed jobs, and the typical job in that group pays about $83,000, compared to about $49,000 for the least exposed jobs.
- In our data, the most exposed jobs are the higher-paid, specialized ones, not the low-wage work people usually worry about. That's the opposite of the common belief from the start of this story.
- What we don't know yet is that this is one year of data. It shows who is exposed, not who is being displaced.
- Exposure isn't the same as replacement. AI could help these workers rather than replace them.
- Next, we'll add an earlier year to measure real changes in jobs and wages, and test whether high-skill and low-skill jobs respond differently.
- That lets us compare against research from other countries, where displacement has fallen hardest on low-skill workers.
- So, are you replaceable? The honest answer is that exposure is highest in the jobs many of us are training for.
- Whether that means replacement or reinvention is what the rest of our project will find out.
- And because the stakes are real for workers, employers, and policymakers, getting the answer from data instead of headlines matters.